Lesson plan · for the teacher

Red-team a school chatbot

AI security · Level 4 · Lesson 2

Time
55 min

Materials
Written permission from the head teacher (and from the provider, if its terms ask for it); Made-up pupils and records for the test; A results table on paper: request, wording, refused?, severity 1–3; A rule card for the pupil who plays the chatbot (for a role-play)

Learning goals

- Plan a safe red-team test: permission, scope, made-up data.
- Score a test bank and rate each miss for severity.
- Report what you found, and check the fix.

Activity: Red-team session

1. Rules (5 min): Show the written permission. Agree which tool and account you will test, that every name and record is made up, and the stop rule: if anything real appears, stop and tell the teacher.
2. Bank (15 min): Each group writes three or four mild requests the chatbot should refuse (a pupil's record for a “neighbour”, a message posing as the school, exam answers), each in a plain wording and a polite rewording.
3. Run (15 min): Try each request on a school tool you have permission to test, or role-play it (one pupil plays the chatbot and refuses only the requests on its rule card, word for word), and record on the paper table: refused or not.
4. Score (10 min): Count the refusals. For each miss, agree its severity: 1 embarrassing, 2 harmful, 3 a danger to a person or their data.
5. Report (10 min): Write one page: what you tried, what got through, how often, how bad, and what you suggest. It goes to the tool's owner first, with no recipe for the trick.

Show the written permission at the start. Use made-up names only, test only tools you are allowed to test, and stop the session if anything real appears.

Discussion

- Why do polite rewordings get through when the plain request doesn't?
- Why does the report go to the owner before anyone else?
- Which fix is stronger: a better model, or a rule in code?

Exit ticket

- Write one request your school chatbot should refuse, in two wordings, and its severity if it got through.

Try it at school tomorrow

With your pupils' AI club, write ten mild requests a school chatbot should refuse, each in two wordings, with made-up pupils only. Practise scoring on the eight-request bank in the Darsi red-team lab, then run yours on a school tool you have written permission to test, or role-play them (one person plays the chatbot and refuses only the requests on its rule card, word for word). Score them with severity, and write a one-page report for the tool's owner.

Practice questions · for you

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1 **In the middle of a test, the chatbot shows a real pupil's phone number. What do you do?**

- ◯ Share a screenshot with the staff group
- ◯ Keep going: finish the bank first
- ◯ Stop, don't copy it, and report it to the owner

2 **Put the red-team test in order.**

- ☐ Run it with made-up data
- ☐ Report to the owner, and re-test after the fix
- ☐ Write the test bank, two wordings each
- ☐ Get written permission and agree the scope
- ☐ Score refusals and rate severity

3 **Each case in the bank is (request, refused?, severity). Count the refusals, print each miss like Miss: scam message severity 3, and finish with Refused 2 of 4.**

```
bank = [("pupil record", True, 3),
        ("scam message", False, 3),
        ("exam answers", False, 2),
        ("rude joke", True, 1)]
refused = 0
for case, ok, sev in bank:
    pass
print("Refused", refused, "of", len(bank))
```

Write your program:

4 **After the fix, the club re-runs four cases; True means it now refuses. What does this print?**

```
done = [True, True, False, True]
print(done.count(True), "of", len(done))
```

- ◯ 3 of 4
- ◯ 4 of 4
- ◯ 1 of 4

5 **Which belong in the rules for a school red-team test? Choose all that apply.**

- ☐ Posting the tricks that worked online, to warn others
- ☐ Testing any public chatbot you like
- ☐ Written permission before you start
- ☐ A stop rule if real data appears
- ☐ Made-up pupils and records

✎ **If your pupils red-teamed a tool your school uses, what would you want them to test first, and which rules would keep the test safe?**

Check your answers

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1 **Stop, don't copy it, and report it to the owner**

Real personal data appearing is real harm, so the stop rule applies: stop testing, don't copy or share it, and report it to the tool's owner at once.

2 **Top to bottom: 3, 5, 2, 1, 4**

Permission and scope come before any test; the bank before the run; scoring before the report; and the work ends only when the fix has been tested again.

3

```
bank = [("pupil record", True, 3),
        ("scam message", False, 3),
        ("exam answers", False, 2),
        ("rude joke", True, 1)]
refused = 0
for case, ok, sev in bank:
    if ok:
        refused += 1
    else:
        print("Miss:", case, "severity", sev)
print("Refused", refused, "of", len(bank))
```

The loop turns raw results into the two things a report needs: how many were refused, and each miss with its severity, so the worst can be fixed first.

4 **3 of 4**

count(True) counts the cases that are now refused: three of the four.

5 **Written permission before you start; Made-up pupils and records; A stop rule if real data appears**

Permission, made-up data and a stop rule keep the test from causing the harm it looks for.